

Erdős Problem #301

EP-301

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $f(N)$ be the size of the largest $A \subseteq \{1, \dots, N\}$ such that there are no solutions to $\frac{1}{a} = \frac{1}{b_1} + \dots + \frac{1}{b_k}$ with distinct $a, b_1, \dots, b_k \in A$?

Estimate $f(N)$. In particular, is it true that $f(N) = (\frac{1}{2} + o(1))N$?